TED UNIVERSITY

Department of Computer Engineering | CMPE 491

SENIOR DESIGN PROJECT I
PROJECT PROPOSAL

ReviewShield: A Multi-Signal Framework for Detecting Coordinated Fake Reviews in Turkish E-Commerce

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Academic year	2026–2027
Submission date	16 October 2026
Project webpage	https://aliyagizt.github.io/ReviewShield/

1. Executive Summary

ReviewShield is a proposed platform-independent software prototype for identifying suspicious, potentially coordinated reviewing activity in Turkish e-commerce data. Its primary unit of analysis is a group of reviewer accounts rather than a single review. A review may appear ordinary in isolation while a cluster of accounts repeatedly posts similar text about the same products within unusually short intervals. The system will examine these complementary signals and prioritize groups for human review.

The proposed workflow accepts structured review data (CSV or JSON), performs preprocessing, extracts text-similarity and temporal features, constructs a reviewer–product graph, detects candidate communities, and produces an interpretable 0–100 suspicion score. A simple web dashboard will show flagged groups, supporting evidence, and basic visualizations. The score is a prioritization measure, not proof of fraud or a calibrated probability of dishonesty.

The project will compare single-signal baselines with a combined approach and report performance and limitations on datasets with suitable fields. If Turkish ground-truth labels are insufficient, controlled synthetic coordinated groups will be inserted into a separate Turkish review background for limited experimental evaluation. Findings on real and synthetic data will be reported separately.

2. Motivation and Problem Definition

Product ratings and customer feedback influence purchasing decisions. Review manipulation can involve multiple accounts acting in a coordinated manner, for example by posting templated praise, targeting the same products, or creating synchronized review bursts. Text-only methods can miss such activity because individual reviews may be fluent and plausible. Conversely, an unusual surge in legitimate reviews during a promotion can resemble manipulation. A reliable prototype must therefore consider several weak signals together and avoid treating any one indicator as conclusive.

The central problem is to detect groups of accounts exhibiting unusually coordinated behavior from available review text, timestamps and reviewer–product interactions. The work focuses on suspicious patterns rather than attributing intent or making automated enforcement decisions.

Research questions

- Does combining text, temporal and graph signals improve detection of coordinated groups compared with individual signals?
- How well does the approach recover controlled groups under different levels of textual similarity and temporal coordination?
- Which signals contribute to a flagged result, and what common legitimate behaviors generate false alarms?

3. Objectives and Scope

The project will deliver a reproducible analysis pipeline, a basic user-facing demonstration and an empirical comparison of approaches. The minimum viable scope is intentionally limited to batch analysis of uploaded datasets; real-time platform integration, scraping, graph neural networks and automated moderation are excluded.

- Process review records with review ID, anonymized account ID, product ID, text, rating and timestamp, when available.
- Compute Turkish sentence-level semantic similarity using a suitable pretrained sentence-embedding model.
- Identify review bursts with a transparent rolling-window or statistical threshold method.
- Construct a weighted account–account graph based on shared products, text similarity and temporal proximity; identify candidate communities.
- Produce group-level suspicion scores with clear explanations and rank groups for analyst inspection.
- Compare text-only, time-only, graph-only and combined configurations on documented evaluation datasets.
- Present results in a modest web interface with review summaries, suspicious groups and a graph view.

4. Proposed Methodology

4.1 Data acquisition and preprocessing

The input schema will include review_id, account_id, product_id, review_text, rating and timestamp. Additional metadata will be optional. The team will first inspect publicly available or otherwise authorized datasets for field completeness and licensing. Missing timestamps or account identifiers will limit the corresponding analysis and will be explicitly documented rather than inferred. Personally identifying data will not be required. Turkish text will be normalized conservatively so that meaningful spelling, negation and sentiment cues are not unintentionally removed.

4.2 Text similarity

A Turkish-compatible or multilingual sentence-embedding model will transform each review into a vector. Cosine similarity will identify semantically related comments about a product or across a candidate group. The team will validate the selected model on a small set of manually inspected Turkish examples before choosing similarity thresholds. Near-duplicate indexing (such as MinHash) is a possible later optimization, not a prerequisite for the prototype.

4.3 Temporal anomaly signals

Review activity will be summarized by product and time window. Rolling counts and deviations from a local baseline will identify bursts; repeated near-simultaneous activity across accounts will provide a separate coordination signal. A burst alone will not be classified as fraud because promotions and viral events can produce legitimate peaks. Windows and thresholds will be specified and sensitivity-tested.

4.4 Reviewer–product graph and candidate groups

A bipartite graph will connect anonymized accounts to reviewed products. A projected account graph will connect pairs that review common products. Edge weights will incorporate evidence from co-reviewed products, review timing and text similarity, with attention to reducing the influence of extremely popular products. A standard community-detection method, initially Louvain or another feasible alternative, will identify candidate groups. Graph construction and threshold choices will be recorded to support reproducibility.

4.5 Fusion, scoring and explanations

Candidate groups will receive normalized text, temporal and structural feature scores. The initial fusion method will be a transparent weighted combination, producing a 0–100 ranking score. The dashboard will display the underlying signals, representative similar comments, overlapping products and temporal patterns. Learned fusion models and SHAP explanations are stretch goals only if reliable labeled training data and sufficient time are available.

5. System Overview and Technology Choices

The pipeline will consist of data import, validation and preprocessing; three feature-analysis modules; candidate-group generation and risk scoring; and a results API and web interface. A batch-oriented design will simplify testing and keep the prototype manageable for a four-person team.

Component	Proposed implementation
Data processing	Python, pandas, NumPy
Text analysis	Sentence Transformers / Hugging Face
Temporal features	pandas and SciPy or simple statistical rules
Graph analysis	NetworkX and one community-detection algorithm
Backend	FastAPI
Frontend	React; optional graph visualization library
Storage	SQLite initially; PostgreSQL if needed
Evaluation	scikit-learn and reproducible notebooks

These tools are provisional choices rather than claims of completed implementation. Selection will depend on available datasets, team familiarity and prototype performance.

6. Evaluation Strategy

The experimental design will separate detection capability from interface development. First, the team will establish simple baselines using only text similarity, only review timing and only graph structure. The combined method will then be evaluated using the same data splits and metrics. A small ablation study will help determine whether each signal provides useful incremental evidence.

If an appropriately licensed labeled dataset with reviewer, product and time fields can be found, it will be used to assess generalization; label provenance and limitations will be documented. For Turkish experiments, a generator may inject controlled groups into a background of real, legally obtained reviews. Scenarios will include identical comments, paraphrased comments and less synchronized posting. Synthetic results will not be represented as real-world fraud detection accuracy.

Where labels permit, evaluation will report precision, recall, F1 and precision at top-k flagged groups; group recovery may be measured by overlap with injected membership. Runtime and false-positive examples will also be reported. ROC-AUC will only be used where labels and score distributions make it meaningful. Evaluation will emphasize a reproducible protocol and candid error analysis rather than a predetermined performance target.

7. Innovation, Feasibility and Risks

The proposed contribution is an explainable, group-oriented combination of semantic, temporal and relational evidence for Turkish product reviews. The project will test this integration experimentally; it does not claim to be the first such method or guarantee superior performance. Its engineering value lies in designing an end-to-end, auditable prototype and systematically comparing the signals.

- Data availability: Complete Turkish datasets with account and time fields may be scarce. Mitigation: perform an early dataset audit and separate real-data and synthetic-data claims.
- False positives: Promotions, product launches and legitimate customer communities may resemble coordination. Mitigation: show evidence, permit human review and analyze errors.
- Model limitations: Semantic similarity does not imply deception. Mitigation: combine independent evidence and avoid binary accusations.
- Graph size: Dense account projections may become expensive. Mitigation: restrict candidate edges using product overlap and thresholding.
- Privacy and ethics: Use authorized datasets and anonymized identifiers; avoid live scraping and public accusations.

8. Work Plan and Expected Deliverables

CMPE 491 (Fall 2026) will emphasize proposal, literature and requirements, analysis, high-level design, and an initial demonstrable prototype. CMPE 492 (Spring 2027) is intended for further implementation, integration and experimental evaluation. This division follows the two-semester senior design structure; exact second-semester milestones will be confirmed with the supervisor.

Stage	Planned output
Early CMPE 491	Literature review, dataset feasibility, project specifications and backlog
Mid CMPE 491	Requirements and analysis report; preliminary text/time baseline
Late CMPE 491	High-level design, initial graph demonstration, project webpage and presentation
CMPE 492	Integrated pipeline, dashboard, ablation experiments, error analysis and final demonstration

The expected final artifacts are a documented analysis pipeline, a basic web dashboard, source code, an evaluation dataset protocol, comparative experiment results, and technical reports. Group members will share responsibility for integration and documentation; specific task allocation will be recorded in the project backlog.

9. Preliminary Literature Review

Ott et al. (2011) studied deceptive opinion spam using linguistic features and supervised learning, providing a foundation for text-based review analysis. Mukherjee et al. (2012) investigated fake reviewer groups and showed the relevance of group-level patterns. Rayana and Akoglu (2015) explored collective opinion spam detection by combining network and metadata evidence. These works motivate the proposed comparison of text-only and multi-signal group-oriented approaches. The subsequent Project Specifications / Literature Review submission will expand this initial review and critically examine comparable approaches and available datasets.

10. References

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